

# WMO First Mile Data Management

# World Meteorological Organization

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## **i. Abstract**

WMO regulates the real-time international exchange of observation data to ensure the availability of high-quality observational data for many applications in particular for meteorological and hydrological forecasts, and allowing Prediction Centers (for Numerical Weather Prediction – NWP as well as climate and hydrological prediction) to be able to produce reliable products. National Meteorological and Hydrological Services (NMHSs) use NWP products to monitor and predict the weather and hydrological conditions to issue forecasts and warnings for the protection of the lives and property of citizens against natural hazards, notably in the framework of the Early Warnings for All initiative.

NMHSs operate networks comprising different observing systems to meet the observational data requirements for national, regional and international needs, and to exchange data internationally, in accordance with the WMO technical regulations. Data are collected from the observing platforms, processed and quality controlled before being used nationally and exchanged internationally. WMO does not regulate this initial data collection segment of transmission of data from the observing station to the data collection and processing system (first-mile data collection).

This document defines the encoding and transmission of observation data from remote monitoring stations (Nodes) to a central data collection system (Hub).

## **ii. Keywords**

The following are keywords to be used by search engines and document catalogues.

wmo, observations, first mile, node, hub, protobuf, mqtt

## **iii. Security Considerations**

TODO

No security considerations have been made for this standard.

# Chapter 1. Scope

This document defines the encoding and transmission of observation data from remote monitoring stations (Nodes) to a central data collection system (Hub).

This specification defines the conformance requirements for WMO First Mile Data Management (data model, encoding, transmission). Annex A defines the abstract test suite.

---

[1] <https://community.wmo.int/governance/commission-membership/commission-observation-infrastructures-and-information-systems-infcom/commission-infrastructure-officers/infcom-management-group/standing-committee-information-management-and-technology-sc-imt>

[2] <https://community.wmo.int/governance/commission-membership/infcom>

# Chapter 2. Conformance

Conformance with this standard shall be checked using the tests specified in Annex A (normative) of this document.

WMO First Mile Data Management defines the encoding and transmission of observation data from remote monitoring stations (Nodes) to a central data collection system (Hub).

This standard identifies numerous Requirements Classes which define the functional requirements.

The mandatory Requirements Classes for this specification are:

- "Data Model"
- "Protocol"

# Chapter 3. References

- Google LLC: Protocol Buffers Documentation (2026) <sup>[1]</sup>

[1] <https://protobuf.dev>



# Chapter 4. Terms and definitions

This document uses the terms defined in [OGC Policy Directive 49](#), which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word “shall” (not “must”) is the verb form used to indicate a requirement to be strictly followed to conform to this Standard and OGC documents do not use the equivalent phrases in the ISO/IEC Directives, Part 2.

This document also uses terms defined in the OGC Standard for Modular specifications ([OGC 08-131r3](#)), also known as the 'ModSpec'. The definitions of terms such as standard, specification, requirement, and conformance test are provided in the ModSpec.

The following additional terms and definitions also apply.

## 4.1. Node

Remote station or system that collects observation data and transmits it to a Hub. A Node comprises a Node Device and optionally one or more Observer Devices. Communication is strictly unidirectional; the Node transmits data to the Hub but does not receive data from it.

## 4.2. Hub

Software system at the central server that receives Data and Metadata messages from one or more Nodes, decodes the Protobuf payloads, and prepares the observation data for further processing and distribution. Communication is strictly unidirectional; the Hub receives data from Nodes but does not transmit data to them.

## 4.3. Protocol

The IP based set of rules that determine how data is transferred from the Node to the Hub.

## 4.4. Data Format

A data format is the organisation of information according to specifications. It defines how the data is structured and encoded.

## 4.5. Data Serialization

The process of converting structured data into a Data Format that can be easily transmitted and reconstructed later.

## 4.6. Data Schema

A formal description in the declarative language defining the structure and constraints for the Data Format.

## 4.7. Observation

A single set of one or more measured values captured at a specific point in time, referencing a `ParameterDefinition`. Observations are transmitted within Data messages.

## 4.8. Metadata

A message type that provides configuration and contextual information about a Node, including descriptions of the Node Device, Observer Devices, parameter definitions, and namespace mappings. The Metadata message provides the necessary context for interpreting Data messages.

## 4.9. Cell method

Statistical method used to aggregate measurements over a cell period. Specified using the `CellMethod` enumeration. Examples include `POINT` (instantaneous value), `MEAN`, `MAXIMUM`, `MINIMUM`, and `STANDARD_DEVIATION`.

## 4.10. Cell period

Duration in seconds over which measurements are aggregated using the cell method. A value of zero indicates an instantaneous (point) measurement with no aggregation interval.

## 4.11. Empty value

Explicit representation of a missing or unavailable measurement. Missing values shall be encoded using the `emptyValue` field rather than omitted, represented as zero, or represented as false. This ensures a clear distinction between a measurement that could not be obtained and a measurement with a value of zero.

## 4.12. First mile

The transmission path from a remote observation site to the central data collection infrastructure. In bandwidth-constrained environments such as satellite links, the first mile refers to the path from the satellite ground base station to the Hub.

## 4.13. Namespace

Declaration that associates a short key with a URI identifying an external naming convention vocabulary. Namespaces are declared in the `Metadata.namespaces` field and referenced in `Parameter.standardNames` entries. Example vocabularies include the CF Metadata Conventions and the WMO First Mile parameter registry.

## 4.14. Node device

The primary device within a Node, typically a datalogger or observation station controller,

responsible for acquiring data from Observer Devices and transmitting encoded payloads to the Hub. Described by the **Node** message in the data model.

## 4.15. Observer device

An individual sensor or measurement device connected to a Node Device. Observer Devices are described by **ObserverDevice** messages within the **Metadata** message. Each Observer Device shall be assigned an identifier that is unique within the context of its Node Device.

## 4.16. Parameter definition

A named grouping of one or more related Parameters that share a common identifier and description. Parameter Definitions are declared in **Metadata** messages and referenced by identifier in **Observation** messages.

## 4.17. Standard name

A controlled vocabulary entry from an external naming convention that unambiguously identifies a measured parameter. Standard names are associated with Parameters through namespace-keyed mappings in the **Parameter.standardNames** field.

## 4.18. Abbreviated terms

Table 1. Symbols and abbreviated terms

| Abbreviation | Term                                                               |
|--------------|--------------------------------------------------------------------|
| CF           | Climate and Forecast (CF) Metadata Conventions                     |
| INFCOM       | Commission for Observation, Infrastructure and Information Systems |
| Protobuf     | Protocol Buffers                                                   |
| SC-IMT       | Standing Committee on Information Management and Technology        |
| HMEI         | HydroMeteorological and Environmental Industry                     |
| MQP          | Message Queuing Protocol                                           |
| MQTT         | Message Queuing Telemetry Transport                                |
| PubSub       | Publish / Subscribe                                                |
| QoS          | Quality of Service                                                 |
| URI          | Uniform Resource Identifier                                        |
| URL          | Uniform Resource Locator                                           |
| WIS          | WMO Information System                                             |
| WMO          | World Meteorological Organization                                  |

# Chapter 5. Conventions

This section provides details and examples for any conventions used in the document. Examples of conventions are symbols, abbreviations, use of JSON Schema, or special notes regarding how to read the document.

## 5.1. Identifiers

The normative provisions in this Standard are denoted by the URI:

<http://wis.wmo.int/spec/fmdm/1>

All requirements and conformance tests that appear in this document are denoted by partial URIs which are relative to this base.

## 5.2. Examples

Protocol topic examples provided in this specification are encoded as **plain text strings**.

Data and metadata message examples provided in this specification are encoded as JSON.

Complete examples can be found at <https://schemas.wmo.int/fmdm/1.0.0/examples>

## 5.3. Schemas

Data and metadata message schemas can be found at <https://schemas.wmo.int/fmdm/1.0.0>

## 5.4. Schema representation

**Protobuf** is used throughout this standard to define the structure of data and metadata messages.

Data and metadata message instances are always defined as JSON.

# Chapter 6. Overview (informative)

## 6.1. Background

In February 2024, a WMO workshop in Geneva brought together representatives from WMO and the Hydro-Meteorological and Environmental Industry (HMEI) to address the lack of standardization in first-mile data collection. While international data exchange is standardized through WMO, no equivalent standardization exists for the first-mile segment, the connection between field equipment and centralized data collection systems, leading to a proliferation of incompatible proprietary formats and significant integration effort for operators of observing networks. The workshop recommended focusing first on output interoperability, and the Standing Committee on Information Management and Technology (SC-IMT) subsequently established the Task Team on First-mile Data Collection Standardization (TT-1M) to produce this specification.

[TODO] Add data flow figure showing 1st mile, DCPS and WIS2.

## 6.2. Solution overview

The first-mile segment covers the connection between field equipment and the systems that collect and process observations. In this specification, the field equipment is represented as the Node and the receiving system as the Hub. Communication is strictly unidirectional, with the Node transmitting protobuf payloads over MQTT to the Hub. The Node represents the remote station or logger through a Node Device and may describe attached sensors as Observer Device entries, as illustrated in [Figure 1](#). The two top-level payloads are the **data** message, which carries Observation content, and the **metadata** message, which carries the descriptive context needed to interpret those observations. Detailed message structure is described in Clause 7, standard name binding in Clause 8, and MQTT protocol binding in Clause 9.

[Node and Hub architecture] | [../images/firstmile-archi.drawio.png](#)

*Figure 1. Node and Hub architecture*

### 6.2.1. Message flow

In typical operation, the **metadata** message is sent when the Node is initialized and whenever the station configuration changes, and the **data** message is sent as observations become available. This separates relatively stable descriptive information from frequently changing measurement values and supports both individual and batched Observation delivery. Where a measurement is unavailable, the value is represented explicitly using **emptyValue** rather than being inferred from zero or omission.

### 6.2.2. Conformance class summary

| Conformance class | Summary                                                                                                                                                                                   |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Core Data Model   | Covers the protobuf structure and semantics of the <b>data</b> message, <b>metadata</b> message, Observation content, devices, Parameter Definition content, and explicit missing values. |

| Conformance class     | Summary                                                                                                                   |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------|
| Standard Name Binding | Covers how Parameter Definition entries associate parameters with controlled standard-name vocabularies and namespaces.   |
| MQTT Protocol Binding | Covers how protobuf payloads are conveyed over MQTT in the unidirectional exchange between Node and Hub.                  |
| Node                  | Covers the sender-side behavior for constructing and publishing <b>metadata</b> message and <b>data</b> message payloads. |
| Hub                   | Covers the receiver-side behavior for accepting, decoding, and interpreting payloads received from the Node.              |

# Chapter 7. Data Model

The data model defines the structure and organization of information transmitted from the Node to the Hub using Protocol Buffers (Protobuf) as the Data Format. The data model is formally specified in the protobuf schema `firstmile.proto`.

The data model supports two primary use cases:

- **Transmission of observation data:** The Node transmits measurement values collected from sensors
- **Transmission of metadata:** The Node transmits information about the monitoring site, devices, and parameter definitions

These use cases are supported by two top-level message types: the Data message containing observations, and the Metadata message containing configuration and contextual information. Both messages are defined in a single Protobuf schema and transmitted as independent payloads. The Metadata message provides the necessary context for interpreting the Data message, including descriptions of devices and parameters.

## 7.1. Requirements Class "Data Model"

### 7.1.1. Overview

This Requirements Class provides requirements for the Data Model.

The data model organizes information hierarchically as follows:

- **Messages:** Top-level containers (Data and Metadata) that are transmitted as independent payloads
- **Devices:** Descriptions of physical equipment including the Node Device (node) and Observer Devices (sensors)
- **Observations:** Measurement data points consisting of a parameter identifier, timestamp, and one or more values
- **Parameter definitions:** Metadata describing what is being measured, including units, aggregation methods, and standard name mappings. Parameter definitions also describe how observations are structured in Data messages.
- **Values:** The actual measurement data, supporting multiple numeric types, strings, booleans, and explicit representation of missing data

Each Observation references a ParameterDefinition by identifier. Each ParameterDefinition contains one or more Parameter specifications that describe the measured variable, its unit, temporal aggregation method, and which device performs the measurement.

|                                                                                                           |
|-----------------------------------------------------------------------------------------------------------|
| <b>Requirements Class</b>                                                                                 |
| <a href="http://wis.wmo.int/spec/fmdm/1/req/data-model">http://wis.wmo.int/spec/fmdm/1/req/data-model</a> |

|                |                                                           |
|----------------|-----------------------------------------------------------|
| Target type    | Message encoding                                          |
| Dependency     | <a href="#">Protobuf</a>                                  |
| Pre-conditions | The event message conforms to the Protobuf specification. |

### 7.1.2. Design principles

The data model is designed according to the following principles:

- **Efficiency:** Optimized for bandwidth-constrained environments but focusing on IP communication rather than extremely low-level binary formats
- **Extensibility:** Support for multiple value types and flexible parameter definitions
- **Self-description:** Metadata provides complete context for interpreting observations
- **Explicit missing values:** Clear distinction between unavailable data and zero/false measurements
- **Batch transmission:** Multiple observations may be included in a single Data message
- **Standard name interoperability:** Mapping to external naming conventions through namespace definitions

### 7.1.3. FirstMileMessage

A FirstMileMessage message denotes whether the content describes data or metadata.

| Requirement 1 | /req/data-model/first_mile_message                                            |
|---------------|-------------------------------------------------------------------------------|
| A             | A FirstMileMessage message SHALL be used as the root definition of a message. |
| B             | A FirstMileMessage message SHALL declare <b>data</b> or <b>metadata</b> .     |

### 7.1.4. Location

The Location message specifies the geographic position of a device (either Node or Observer).

| Requirement 2 | /req/data-model/location                                                                                                     |
|---------------|------------------------------------------------------------------------------------------------------------------------------|
| A             | A Location message SHALL define a <b>heightMetres</b> element in metres, above the <b>referenceSurface</b> .                 |
| B             | A Location message SHALL define a <b>referenceSurface</b> element using the ReferenceSurface enumeration defined in Annex B. |

| Recommendation 1 | /rec/data-model/location                                                                                                                                                       |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A                | A Location message SHOULD provide a <b>latitude</b> element describing the latitude in decimal degrees using the WGS84 coordinate system and datum. // TODO validate precision |



|   |                                                                                                                                                                                  |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B | A Location message SHOULD provide a <b>longitude</b> element describing the longitude in decimal degrees using the WGS84 coordinate system and datum. // TODO validate precision |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                     |                                                                                                      |
|---------------------|------------------------------------------------------------------------------------------------------|
| <b>Permission 1</b> | <b>/per/data-model/location</b>                                                                      |
| A                   | A Location message MAY omit the <b>latitude</b> and <b>longitude</b> elements when they are unknown. |

### 7.1.5. Metadata

A Metadata message provides configuration and contextual information about the Node, including descriptions of devices and the parameters they measure. This message shall be transmitted at least once after Node initialization and whenever the configuration changes.

|                      |                                                                                                                                                                 |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Requirement 3</b> | <b>/req/data-model/metadata</b>                                                                                                                                 |
| A                    | A Metadata message SHALL define a <b>node</b> element to describe the Node device (datalogger or station).                                                      |
| B                    | A Metadata message SHALL define zero or more <b>observers</b> to describe individual sensor devices connected to the Node.                                      |
| C                    | A Metadata message SHALL define one or more <b>parameterDefinitions</b> to define the parameters used in observations.                                          |
| D                    | A Metadata message SHALL define a <b>namespaces</b> element to define a mapping of namespace prefixes to their corresponding URIs for standard name resolution. |

### 7.1.6. Metadata.Node

A Node message describes the Node device itself, typically a datalogger or observation station.

|                      |                                                                                                                |
|----------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Requirement 4</b> | <b>/req/data-model/node</b>                                                                                    |
| A                    | A Node message SHALL define a <b>name</b> element describing the name or model designation of the Node device. |

|                         |                                                                                                                                      |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Recommendation 2</b> | <b>/rec/data-model/node</b>                                                                                                          |
| A                       | A Node message <b>name</b> SHOULD contain a human-readable identification of the device, such as its model name or type designation. |
| B                       | A Node message SHOULD define a <b>location</b> element to describe the geographic location of the Node device.                       |

|                     |                             |
|---------------------|-----------------------------|
| <b>Permission 2</b> | <b>/per/data-model/node</b> |
|---------------------|-----------------------------|

|   |                                                                                                                                                                          |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A | A Node message MAY define a <b>url</b> element providing a URI reference to additional metadata about the device.                                                        |
| B | A Node message MAY define a <b>serialNumber</b> element describing the manufacturer's serial number of the device.                                                       |
| C | A Node message MAY define a <b>firmwareVersion</b> element describing the firmware version of the device, typically in format such as <b>5.1</b> or <b>2.7.1-alpha</b> . |

### 7.1.7. Metadata.ObserverDevice

An ObserverDevice message describes an individual sensor or measurement device connected to the Node Device. A Node that measures only its own internal parameters (such as supply voltage or internal temperature) may transmit a Metadata message with no ObserverDevice entries.

Where a sensor is defined as an ObserverDevice but is not currently connected or is otherwise unavailable, the corresponding parameter values in the Data message is represented using **emptyValue** rather than being omitted. This ensures that the Hub can distinguish between a missing sensor reading and the absence of a parameter definition, and that the alignment between values and parameter definitions is maintained.

|                      |                                                                                                                                |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Requirement 5</b> | <b>/req/data-model/observer_device</b>                                                                                         |
| A                    | An ObserverDevice message SHALL define an <b>id</b> element.                                                                   |
| B                    | An ObserverDevice message SHALL define a <b>name</b> element to describe the name or model designation of the observer device. |

|                         |                                                                                                                              |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Recommendation 3</b> | <b>/rec/data-model/observer_device</b>                                                                                       |
| A                       | An ObserverDevice message SHOULD define a <b>location</b> element describing the geographic location of the observer device. |

|                     |                                                                                                                              |
|---------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Permission 3</b> | <b>/per/data-model/observer_device</b>                                                                                       |
| A                   | An ObserverDevice message MAY define a <b>url</b> element providing a URI reference to additional metadata about the device. |
| B                   | An ObserverDevice message MAY define a <b>serialNumber</b> element describing manufacturer's serial number of the device.    |
| C                   | An ObserverDevice message MAY define a <b>firmwareVersion</b> element describing firmware version of the device.             |

#### IMPORTANT

Observer device identifiers shall remain stable across Node restarts and metadata updates to ensure consistent parameter-to-device associations.

### 7.1.8. Metadata.ParameterDefinition

A ParameterDefinition groups one or more related measurement parameters that share common context. Each parameter definition has a unique identifier that is referenced by observations.

Parameters are typically grouped when they are measured at the same timestamp, allowing multiple values to share a single timestamp in observations and thus reducing payload size.

The parameters field contains one or more Parameter messages describing individual measurable variables. When multiple parameters are grouped in a single definition, each observation referencing this definition shall provide values in the same order as the parameters are defined.

| Requirement 6 | /req/data-model/parameter_definition                                               |
|---------------|------------------------------------------------------------------------------------|
| A             | A ParameterDefinition message SHALL define an <b>id</b> element.                   |
| B             | A ParameterDefinition message SHALL define a <b>description</b> element.           |
| C             | A ParameterDefinition message SHALL define one or more <b>parameters</b> elements. |

### 7.1.9. Metadata.ParameterDefinition.Parameter

The Parameter message specifies a single measurable variable, including its name, unit of measurement, aggregation method, and associated device.

The standardNames field enables mapping to external naming conventions such as Climate and Forecast (CF) standard names. This allows parameters to be referenced by different standard parameter types according to the target application of the device using the protocol. The Node is not bound to a specific standard but may include mappings to multiple standards as appropriate for the intended use case. The keys in this map shall correspond to namespace prefixes defined in Metadata namespaces.

The device field specifies which device (either the node or a specific observer) performs this measurement.

| Requirement 7 | /req/data-model/parameter                                                                                                                                                                                                                                                                                                     |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A             | A Parameter message SHALL define a <b>longName</b> element describing the name of the parameter (e.g., "Air Temperature", "Wind Speed").                                                                                                                                                                                      |
| B             | A Parameter message SHALL define a <b>unit</b> element using the <a href="#">WMO Register: Code Table C-6: List of units for TDCFs</a> . // TODO verify                                                                                                                                                                       |
| C             | A Parameter message SHALL define a <b>standardNames</b> map which describes zero or more namespace prefix and standard name mappings for the equivalent parameter name in an external vocabulary. // TODO: was "one or more"; changed to "zero or more" to align with /per/data-model/standard_name_binding_assignment Part A |
| D             | A Parameter message SHALL define a <b>device</b> element describing the device (Node or Observer) that measures this parameter.                                                                                                                                                                                               |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E | A Parameter message SHALL define a <code>cellMethod</code> element describing the statistical or aggregation method applied to the measurement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| F | A Parameter message SHALL define a <code>cellPeriodSeconds</code> element describing the period of aggregation in seconds (0 for instantaneous point measurements).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| G | When <code>cellMethod</code> is <code>POINT</code> , the value SHALL be 0, indicating an instantaneous measurement. For all other cell methods, the value SHALL be greater than 0, indicating the duration of the aggregation period. For example: * A 30-second mean temperature would have a <code>cellMethod</code> value of <code>MEAN</code> and a <code>cellPeriodSeconds</code> value of 30. An instantaneous wind direction would have a <code>cellMethod</code> value of <code>POINT</code> and a <code>cellPeriodSeconds</code> value of 0. A 5-minute maximum wind gust would have a <code>cellMethod</code> of <code>MAXIMUM</code> and a <code>cellPeriodSeconds</code> value of 300. |
| H | When multiple parameters are grouped in a single definition, each observation referencing this definition SHALL provide values in the same order as the parameters are defined.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

### 7.1.10. Standard name binding

Standard name binding enables Parameter definitions within a Metadata message to be associated with controlled vocabulary entries from external naming conventions. This allows observation data to be unambiguously interpreted by Hubs and downstream systems without dependence on the Parameter `LongName` element, which is a human-readable label and not guaranteed to be unique or machine-interpretable.

The binding mechanism uses two related fields in the data model:

- Metadata `namespaces`: declares the naming convention vocabularies in use, associating a short namespace key with a URI that identifies the vocabulary
- Parameter `standardNames`: associates each namespace key with the corresponding standard name for the parameter within that vocabulary

Standard name binding follows a three-level hierarchy of vocabularies, described in [Standard name hierarchy](#). The use of standard names is optional but recommended where machine-to-machine interoperability is required. A Node may bind a single parameter to standard names from multiple vocabularies simultaneously.

#### 7.1.10.1. Metadata namespace declaration

The Metadata message `namespaces` element declares the naming convention vocabularies used by the Node. It is a map from namespace prefix to namespace URI.

|                      |                                                                       |
|----------------------|-----------------------------------------------------------------------|
| <b>Requirement 8</b> | <b><code>/req/data-model/standard_name_binding_declaration</code></b> |
|----------------------|-----------------------------------------------------------------------|

|   |                                                                                                                                                 |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------|
| A | The Metadata message <b>namespaces</b> element SHALL define zero or more namespace maps.                                                        |
| B | Each namespace map SHALL consist of a namespace prefix and URI.                                                                                 |
| C | A namespace map prefix SHALL be a non-empty string containing only alphanumeric characters and underscores.                                     |
| D | A namespace map URI SHALL be a resolvable, publicly accessible URI that can be dereferenced.                                                    |
| E | The CF namespace prefix SHALL be declared using the value <b>cf</b> .                                                                           |
| F | The CF namespace URI value SHALL be a URI that resolves to a machine-readable vocabulary server endpoint publishing the CF standard name table. |

|                         |                                                                                                                                                                                                                                                                |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Recommendation 4</b> | <b>/rec/data-model/standard_name_binding_declaration</b>                                                                                                                                                                                                       |
| A                       | For CF conventions, the namespace URI value SHOULD be set to the NERC Vocabulary Server mirror of the CF standard name table (collection P07): <a href="https://vocab.nerc.ac.uk/collection/P07/current">https://vocab.nerc.ac.uk/collection/P07/current</a> . |

|                     |                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------|
| <b>Permission 4</b> | <b>/per/data-model/standard_name_binding_declaration</b>                                   |
| A                   | When no standard names are used, the <b>namespaces</b> map MAY be either empty or omitted. |

#### NOTE

If the CF Conventions community publishes an official vocabulary server endpoint in the future, that URI should be used in preference to the NERC mirror. Hubs should be prepared to recognise URIs from recognised CF standard name vocabulary server mirrors as equivalent for the purposes of name resolution.

### 7.1.10.2. Standard name assignment

The Parameter message **standardNames** field maps namespace keys to the standard name for that parameter within the corresponding vocabulary. It is a map from namespace key to standard name value.

|                      |                                                                                                                            |
|----------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Requirement 9</b> | <b>/req/data-model/standard_name_binding_assignment</b>                                                                    |
| A                    | A Node SHALL NOT include a standard name entry for a namespace that is not declared in the corresponding Metadata message. |
| B                    | Each namespace key SHALL have a corresponding key in the Metadata message <b>namespaces</b> set of keys.                   |

|                     |                                                           |
|---------------------|-----------------------------------------------------------|
| <b>Permission 5</b> | <b>/per/data-model/standard_name_binding_assignment</b>   |
| A                   | A parameter MAY carry zero or more standard name entries. |

**NOTE**

Standard name binding is applied in the Parameter message (not the ParameterDefinition message). When a ParameterDefinition message contains multiple Parameter messages, each parameter independently carries its own standard name bindings appropriate for the variable it represents.

### 7.1.10.3. Standard name hierarchy

Standard names are defined at three levels:

- Level 1: CF Metadata Conventions
- Level 2: WMO First Mile parameter registry
- Level 3: Locally managed vocabularies

The Climate and Forecast (CF) Metadata Conventions define the primary vocabulary for standard names in this standard. CF standard names cover the majority of meteorological, hydrological, and oceanographic parameters relevant to first-mile observations. The authoritative list of CF standard names is maintained by the CF Conventions community at <https://cfconventions.org/Data/cf-standard-names/current/build/cf-standard-name-table.html>. CF standard names are lower-case strings with words separated by underscores (e.g., `air_temperature`, `wind_speed`, `relative_humidity`). The CF standard name table at cfconventions.org is the authoritative reference for name validity and definitions, regardless of which vocabulary server URI is used as the namespace identifier.

Parameters that are not covered by the CF standard name vocabulary but are common to first-mile observation deployments are governed by a centrally managed registry maintained by WMO and hosted at <https://codes.wmo.int>.

A locally managed vocabulary is used where a parameter is not covered by either of Levels 1 or 2.

| Requirement 10 | /req/data-model/standard_name_binding_hierarchy                                                                                                                                                                                                                                     |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A              | Standard names SHALL be drawn in the following order, in the following order of precedence: <ul style="list-style-type: none"><li>• Level 1: CF Metadata Conventions</li><li>• Level 2: WMO First Mile parameter registry</li><li>• Level 3: Locally managed vocabularies</li></ul> |
| B              | The CF namespace prefix SHALL be declared using the value <code>cf</code> .                                                                                                                                                                                                         |
| C              | The CF namespace URI value SHALL be a URI that resolves to a machine-readable vocabulary server endpoint publishing the CF standard name table.                                                                                                                                     |
| D              | The WMO First Mile registry namespace prefix SHALL be declared using the value <code>wmo</code> .                                                                                                                                                                                   |
| E              | Locally managed vocabularies SHALL conform to the FAIR Guiding Principles for scientific data management (Findable, Accessible, Interoperable, Reusable). In particular:                                                                                                            |

|   |                                                                                                   |
|---|---------------------------------------------------------------------------------------------------|
| F | The vocabulary SHALL be findable via a persistent, unique identifier (the namespace URI).         |
| G | The vocabulary SHALL be accessible at that URI to all parties participating in the data exchange. |
| H | The vocabulary SHALL use an open, standardised representation (e.g., SKOS/RDF).                   |

|                         |                                                                                                                                                                                          |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Recommendation 5</b> | <b>/rec/data-model/standard_name_binding_hierarchy</b>                                                                                                                                   |
| A                       | The highest vocabulary SHOULD be used when it provides a suitable standard name for a given parameter.                                                                                   |
| B                       | Where a CF standard name exists for a parameter, the Node SHOULD use it in preference to any lower-level vocabulary entry.                                                               |
| C                       | Where a parameter is not covered by CF but is included in the WMO First Mile parameter registry, the Node SHOULD use the registry entry in preference to any locally managed vocabulary. |

|                     |                                                                                                           |
|---------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Permission 6</b> | <b>/per/data-model/standard_name_binding_hierarchy</b>                                                    |
| A                   | Where a parameter is not covered by either of Levels 1 or 2, a Node MAY use a locally managed vocabulary. |
| B                   | Local namespace entries MAY be included alongside Level 1 or Level 2 standard names.                      |

#### NOTE

Parameters that may require Level 2 registry entries include quantities typically measured by data loggers but absent from CF, such as supply voltage, internal device temperature, and station-level pressure (QFF). The registry is managed separately from this standard; refer to <https://codes.wmo.int> for the current list of registered entries.

#### NOTE

A conformant Hub should be capable of resolving namespace URIs as configured for the deployment, including URIs accessible only within a private or organisational network. A conformant Node shall use namespace URIs that are resolvable by the Hub within the operational context of the deployment.

#### IMPORTANT

Where a standard name exists at Level 1 or Level 2, that standard name is to be included in the Parameter message `standardNames` element. Any locally managed entries for the same parameter are supplementary to, not a replacement for, the applicable standard name. // TODO consider making a requirement (but note this is untestable)

### 7.1.11. Metadata.ParameterDefinition.Parameter.DeviceRef

The DeviceRef message associates a parameter with the device that measures it. This message uses a `oneof` field to reference either the node device or a specific observer device.



| Requirement 11 | /req/data-model/device-ref                                                                                                                                                               |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A              | A DeviceRef message SHALL define one and only one of an <b>observerId</b> element as a reference to an ObserverDevice message or a <b>node</b> if the parameter is measured by the Node. |
| B              | When the <b>node</b> element is present, the parameter SHALL be measured by the Node Device itself (such as internal voltage or temperature of a datalogger).                            |
| C              | When the observerId field is set, it SHALL match the id of an ObserverDevice defined in the Metadata message.                                                                            |

### 7.1.12. Data

The Data message is the primary container for transmitting observation data from the Node to the Hub. This message may be transmitted as frequently as needed to deliver current or batched observations.

A Data message may contain multiple Observation messages, enabling efficient batch transmission of measurement data. Each observation is independent and contains its own timestamp and parameter reference.

#### NOTE

The Node may transmit Data messages containing observations that reference parameter definitions not yet received by the Hub. In this case, the Hub shall buffer or queue such observations until the corresponding **Metadata** message is received.

| Requirement 12 | /req/data-model/data                                                                                       |
|----------------|------------------------------------------------------------------------------------------------------------|
| A              | A Data message SHALL define one or more <b>observations</b> elements to describe observation measurements. |

### 7.1.13. Observation

The Observation message represents a single measurement data point collected at a specific time.

| Requirement 13 | /req/data-model/observation                                                                                                 |
|----------------|-----------------------------------------------------------------------------------------------------------------------------|
| A              | An Observation message SHALL define a <b>parameterDefinitionId</b> element using the id of the ParameterDefinition message. |
| B              | An Observation message SHALL define a <b>time</b> element using <b>google.protobuf.Timestamp</b> , in UTC.                  |
| C              | An Observation message SHALL define one or more <b>values</b> elements using the Value message.                             |

#### NOTE

The **parameterDefinitionId** element shall reference a ParameterDefinition **id** from the Metadata message. The number and type of values in the values field shall correspond to the number of parameters defined in the referenced ParameterDefinition.



**NOTE**

The time field uses the `google.protobuf.Timestamp` format, representing time as seconds and nanoseconds since the Unix epoch (1970-01-01T00:00:00Z).

**NOTE**

An observation may contain multiple values when the referenced ParameterDefinition contains multiple parameters. This approach enables grouping values that are measured at the same timestamp, requiring the timestamp to be transmitted only once for multiple values. This reduces payload size, which is particularly beneficial in bandwidth-constrained environments. For example, a parameter definition for "Internal Parameters" might include both supply voltage and internal temperature, requiring two values per observation but sharing a single timestamp.

## 7.1.14. Value

A Value message defines the measurement of data of different types.

| Requirement 14 | /req/data-model/value                                                                                                                                                                             |                       |                                       |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------|
| A              | A Value message SHALL be defined using one of the following data types:                                                                                                                           |                       |                                       |
|                | Field                                                                                                                                                                                             | Type                  | Description                           |
|                | floatValue                                                                                                                                                                                        | float                 | 32-bit floating point number          |
|                | doubleValue                                                                                                                                                                                       | double                | 64-bit floating point number          |
|                | intValue                                                                                                                                                                                          | int32                 | 32-bit signed integer                 |
|                | unsignedIntValue                                                                                                                                                                                  | uint32                | 32-bit unsigned integer               |
|                | int64Value                                                                                                                                                                                        | int64                 | 64-bit signed integer                 |
|                | unsignedInt64Value                                                                                                                                                                                | uint64                | 64-bit unsigned integer               |
|                | stringValue                                                                                                                                                                                       | string                | Text string value                     |
|                | boolValue                                                                                                                                                                                         | bool                  | Boolean true/false value              |
|                | emptyValue                                                                                                                                                                                        | google.protobuf.Empty | Indicates missing or unavailable data |
| B              | When a measurement value is unavailable due to sensor malfunction, communication error, or other causes, the value SHALL be represented using the emptyValue field of type google.protobuf.Empty. |                       |                                       |

The choice of value type should be appropriate for the parameter being measured. Numeric measurements typically use doubleValue or floatValue, while categorical observations may use stringValue. Boolean parameters (such as alarm states) use boolValue.

**NOTE**

The stringValue field is intended for text data only and shall NOT be used to

transmit: - binary data, including base64-encoded binary blobs - serialized objects or data structures - structured data formats such as JSON, XML, or similar encodings

**NOTE**

Binary data transmission and complex structured data are outside the scope of this standard.

**NOTE**

Missing or unavailable measurement values shall NOT be represented as: - numeric zero (0) - NaN (Not a Number) - omission from the values array - any other numeric sentinel value

**NOTE**

The `emptyValue` field shall be used to explicitly and unambiguously indicate missing data.

This approach ensures clear distinction between actual measurements of zero and missing measurements, which is critical for correct data interpretation.

When an Observation references a ParameterDefinition containing multiple parameters, the values in the values array shall be provided in the same order as the parameters are defined in the ParameterDefinition.parameters field. It is not permitted to omit a value from the middle of the array, as this would cause misalignment with the parameter definitions. If a value is unavailable, the `emptyValue` field shall be used at that position to maintain correct alignment.

# Chapter 8. Protocol

TODO one sentence

## 8.1. Requirements Class "Protocol"

### 8.1.1. Overview

This Requirements Class provides requirements for the Protocol.

Unresolved directive in sections/clause\_8\_protocol.adoc -  
include::../requirements/requirements\_class\_protocol.adoc[]

# Annex A: Conformance Class Abstract Test Suite (Normative)

## A.1. Conformance Class: Data Model

**label**

<http://wis.wmo.int/spec/fmdm/1/req/data-model>

**subject**

Requirements Class "Data Model"

**classification**

Target Type:Event Metadata

### A.1.1. Encoding

**label**

/conf/data-model/encoding

**subject**

/req/data-model/encoding

**test-purpose**

Validate that a data model is encoding as Protobuf.

TODO

TODO

## A.2. Conformance Class: Protocol

Unresolved directive in sections/annex\_a\_ats.adoc -  
include::../abstract\_tests/ATS\_class\_protocol.adoc[]

# Annex B: Schemas (Normative)

## NOTE

Schema documents will only be published on [schemas.wmo.int](https://schemas.wmo.int) once the standard has been approved.

## B.1. Data Model

```
syntax = "proto3";

package int.wmo.wis.spec.fmdm.1.conf.data_model;

import "google/protobuf/empty.proto";
import "google/protobuf/timestamp.proto";

enum CellMethod {
  CELL_METHOD_UNSPECIFIED = 0; // Required default
  POINT = 1; // Instantaneous value at a point
  SUM = 2; // Sum over the cell
  MAXIMUM = 3; // Maximum value
  MAXIMUM_ABSOLUTE_VALUE = 4; // Maximum of absolute values
  MEDIAN = 5; // Median value
  MID_RANGE = 6; // Average of maximum and minimum
  MINIMUM = 7; // Minimum value
  MINIMUM_ABSOLUTE_VALUE = 8; // Minimum of absolute values
  MEAN = 9; // Mean (average) value
  MEAN_ABSOLUTE_VALUE = 10; // Mean of absolute values
  MEAN_OF_UPPER_DECILE = 11; // Mean of the upper group of data values defined by
the upper tenth of their distribution
  MODE = 12; // Mode (most common value)
  RANGE = 13; // Absolute difference between maximum and minimum
  ROOT_MEAN_SQUARE = 14; // Root mean square (RMS)
  STANDARD_DEVIATION = 15; // Standard deviation within the cell
  SUM_OF_SQUARES = 16; // Sum of squares
  VARIANCE = 17; // Variance within the cell
}

message DeviceRef {
  oneof target {
    google.protobuf.Empty node = 1; // if field is present => refers to the single
node
    uint32 observerId = 2;
  }
}

message Parameter {
  string longName = 1;
  string unit = 2;
  map<string, string> standardNames = 3;
```

```

DeviceRef device = 4;
CellMethod cellMethod = 5;
uint32 cellPeriodSeconds = 6;
}

message ParameterDefinition {
  uint32 id = 1;
  string description = 2;
  repeated Parameter parameters = 3;
}

message Value {
  oneof kind {
    float floatValue = 1;
    double doubleValue = 2;
    int32 intValue = 3;
    uint32 unsignedIntValue = 4;
    int64 int64Value = 5;
    uint64 unsignedInt64Value = 6;
    string stringValue = 7;
    bool boolValue = 8;
    google.protobuf.Empty emptyValue = 9;
  }
}

message Observation {
  uint32 parameterDefinitionId = 1;
  google.protobuf.Timestamp time = 2;
  repeated Value values = 3;
}

enum ReferenceSurface {
  REFERENCE_SURFACE_UNSPECIFIED = 0; // Default
  MSL = 1;                          // Mean Sea Level
  GEOID = 2;                         // Geoid
  GL = 3;                           // Ground Level
  REFERENCE_ELLIPSOID = 4;          // Reference Ellipsoid
  PRESSURE_1000_HPA = 5;           // 1000 hPa Pressure Level
}

message Location {
  optional double latitude = 1;
  optional double longitude = 2;
  double heightMetres = 3;
  ReferenceSurface referenceSurface = 4;
}

message Node {
  string id = 1;
  string name = 1;
  optional Location location = 2;
}

```

```

optional string url = 3;
optional string serialNumber = 4;
optional string firmwareVersion = 5;
}

message ObserverDevice {
  uint32 id = 1;
  string name = 2;
  optional Location location = 3;
  optional string url = 4;
  optional string serialNumber = 5;
  optional string firmwareVersion = 6;
}

message Data {
  repeated Observation observations = 1;
}

message Metadata {
  Node node = 1;
  repeated ObserverDevice observers = 2;
  repeated ParameterDefinition parameterDefinitions = 3;
  map<string, string> namespaces = 4;
}

message FirstMileMessage {
  oneof content {
    // MUST be published with retained flag
    Metadata metadata = 1;

    // MUST NOT be published with retained flag
    Data data = 2;
  }
}

```

## B.2. Enumerations

### B.2.1. CellMethod

The `cellMethod` field specifies how measurement data is aggregated or processed over the cell period. The `CellMethod` enumeration defines the following values:

| Name                                 | Value | Description                    |
|--------------------------------------|-------|--------------------------------|
| <code>CELL_METHOD_UNSPECIFIED</code> | 0     | Required default               |
| <code>POINT</code>                   | 1     | Instantaneous value at a point |
| <code>SUM</code>                     | 2     | Sum over the cell              |
| <code>MAXIMUM</code>                 | 3     | Maximum value                  |

| Name                   | Value | Description                                                                             |
|------------------------|-------|-----------------------------------------------------------------------------------------|
| MAXIMUM_ABSOLUTE_VALUE | 4     | Maximum of absolute values                                                              |
| MEDIAN                 | 5     | Median value                                                                            |
| MID_RANGE              | 6     | Average of maximum and minimum                                                          |
| MINIMUM                | 7     | Minimum value                                                                           |
| MINIMUM_ABSOLUTE_VALUE | 8     | Minimum of absolute values                                                              |
| MEAN                   | 9     | Mean (average) value                                                                    |
| MEAN_ABSOLUTE_VALUE    | 10    | Mean of absolute values                                                                 |
| MEAN_OF_UPPER_DECILE   | 11    | Mean of the upper group of data values defined by the upper tenth of their distribution |
| MODE                   | 12    | Mode (most common value)                                                                |
| RANGE                  | 13    | Absolute difference between maximum and minimum                                         |
| ROOT_MEAN_SQUARE       | 14    | Root mean square (RMS)                                                                  |
| STANDARD_DEVIATION     | 15    | Standard deviation within the cell                                                      |
| SUM_OF_SQUARES         | 16    | Sum of squares                                                                          |
| VARIANCE               | 17    | Variance within the cell                                                                |

#### NOTE

When cellMethod is set to **POINT**, the measurement represents an instantaneous value and cellPeriodSeconds shall be 0. For all other cell methods, **cellPeriodSeconds** shall specify the duration over which the aggregation is performed.

### B.2.2. ReferenceSurface

| Name                          | Value | Description             |
|-------------------------------|-------|-------------------------|
| REFERENCE_SURFACE_UNSPECIFIED | 0     | Default                 |
| MSL                           | 1     | Mean Sea Level          |
| GEOID                         | 2     | Geoid                   |
| GL                            | 3     | Ground Level            |
| REFERENCE_ELLIPSOID           | 4     | Reference Ellipsoid     |
| PRESSURE_1000_HPA             | 5     | 1000 hPa Pressure Level |

#### NOTE

The choice of reference surface is determined by the user based on the measurement context and requirements. For example, meteorological observations of air temperature typically use height above ground level (GL), while atmospheric pressure observations may reference mean sea level (MSL).



# **Annex C: Examples (Informative)**

## **C.1. Transmission topics**

## **C.2. Data and metadata messages**

# Annex D: Bibliography

- IANA: Link Relation Types, <https://www.iana.org/assignments/link-relations/link-relations.xml>

# Annex E: Revision History

| Date       | Release  | Editor       | Primary clauses modified | Description      |
|------------|----------|--------------|--------------------------|------------------|
| 2026-06-15 | Template | Tom Kralidis | all                      | initial revision |